Felix_PCA_script

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Setup

Load libraries

```r
library(vcfR)
```

```
##
##    *****       ***   vcfR   ***       *****
##    This is vcfR 1.16.0
##      browseVignettes('vcfR') # Documentation
##      citation('vcfR') # Citation
##    *****       *****      *****       *****
```

```r
library(ade4)
library(adegenet)
```

```
##
##    /// adegenet 2.1.11 is loaded ////////////
##
##    > overview: '?adegenet'
##    > tutorials/doc/questions: 'adegenetWeb()'
##    > bug reports/feature requests: adegenetIssues()
```

```r
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(RColorBrewer)
```

Prepare and parse vcf

Read in VCF

```r
vcf <- read.vcfR("/home/FM/sblanco/GWAS_8072026/usub/05_filtering/usub_filtered_miss0_7
5.vcf.gz")
```

```
## Scanning file to determine attributes.
## File attributes:
##   meta lines: 83
##   header_line: 84
##   variant count: 19105
##   column count: 44
## Meta line 83 read in.
## All meta lines processed.
## gt matrix initialized.
## Character matrix gt created.
##   Character matrix gt rows: 19105
##   Character matrix gt cols: 44
##   skip: 0
##   nrows: 19105
##   row_num: 0
## Processed variant 1000Processed variant 2000Processed variant 3000Processed variant 4
000Processed variant 5000Processed variant 6000Processed variant 7000Processed variant 8
000Processed variant 9000Processed variant 10000Processed variant 11000Processed variant
12000Processed variant 13000Processed variant 14000Processed variant 15000Processed vari
ant 16000Processed variant 17000Processed variant 18000Processed variant 19000Processed
variant: 19105
## All variants processed
```

Assign samples

```r
vcf_samples <- colnames(vcf@gt)[-1]

pop.file <- data.frame(
  sample = vcf_samples,
  pop  = substr(vcf_samples, 1, 4),
  stringsAsFactors = FALSE
)

pop.file <- pop.file[match(vcf_samples, pop.file$sample), ]

all(pop.file$sample == vcf_samples) #shoould be all TRUE
```

```
## [1] TRUE
```

```
data.genind <- vcfR2genind(vcf, ploidy = 1, pop = pop.file$pop)
```

Check how much missing data

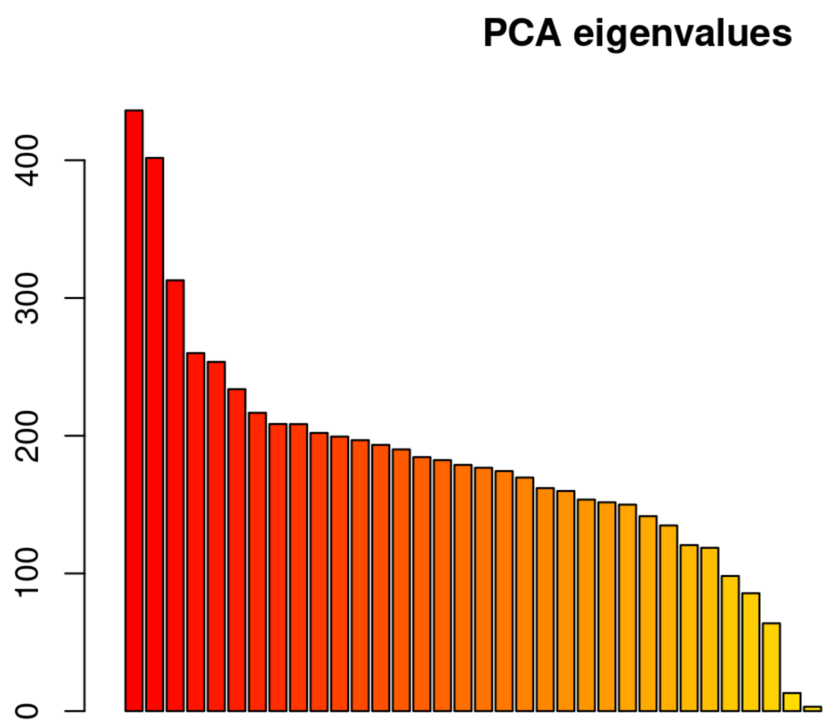
```
sum(is.na(data.genind$tab))
```

```
## [1] 152390
```

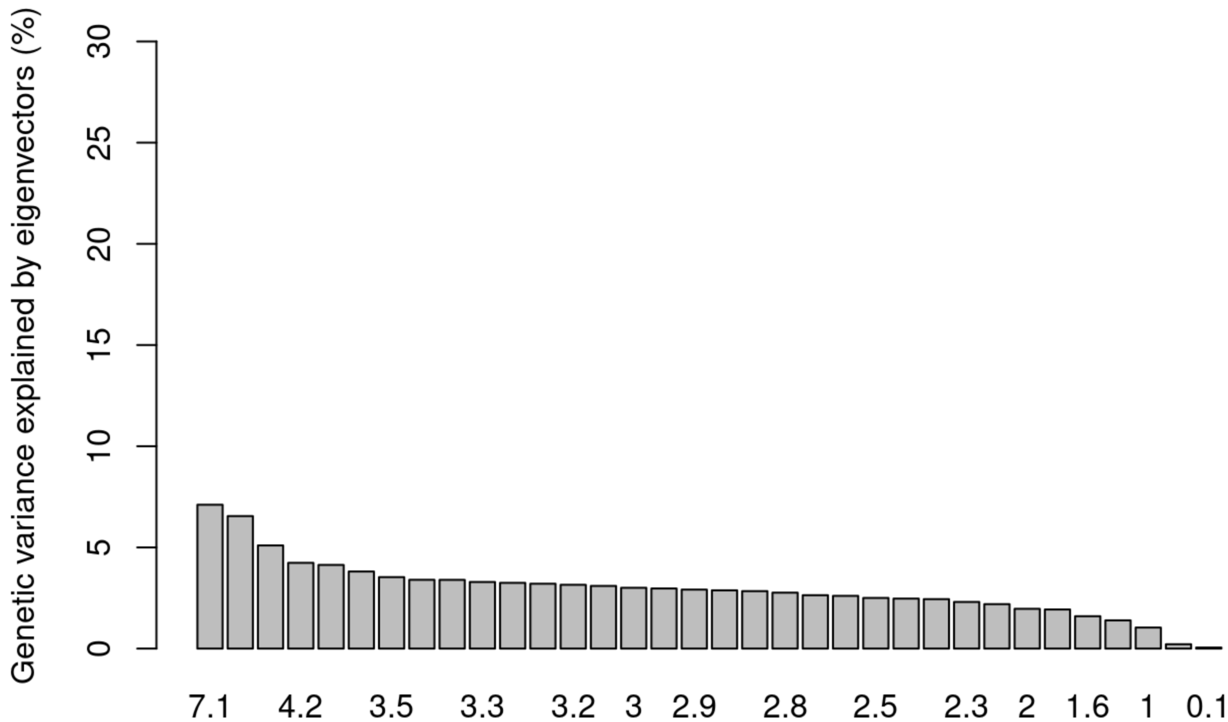
```
X <- tab(data.genind, freq = TRUE, NA.method = "mean")
```

Calculate eigenvalues

```
pca1 <- dudi.pca(X, scale = FALSE, scannf = FALSE, nf = 3)
barplot(pca1$eig[1:50], main = "PCA eigenvalues", col = heat.colors(50))
```



```
percent <- pca1$eig / sum(pca1$eig) * 100
barplot(percent, ylab = "Genetic variance explained by eigenvectors (%)", ylim = c(0, 3
0), names.arg = round(percent, 1))
```



Make PCA

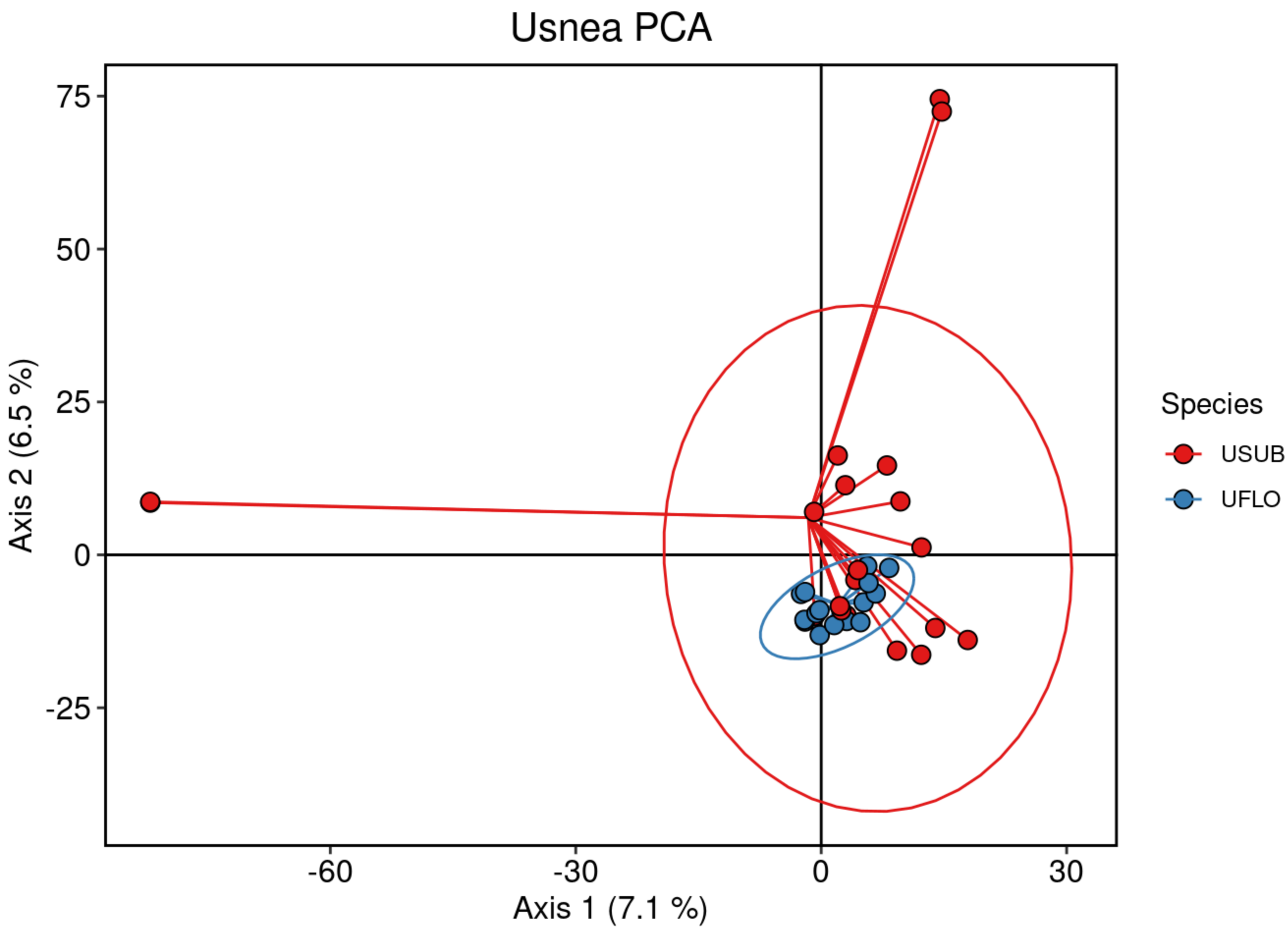
```
ind_coords <- as.data.frame(pca1$li) #Create a data.frame containing individual coordina
tes
colnames(ind_coords) <- c("Axis1", "Axis2", "Axis3") #Rename columns of dataframe
ind_coords$Ind <- indNames(data.genind) #Add a column with the site IDs
ind_coords$Species <- data.genind$pop
centroid <- aggregate(cbind(Axis1, Axis2, Axis3) ~ Species, data = ind_coords, FUN = mea
n) #Calculate centroid (average) position for each population
ind_coords <- left_join(ind_coords, centroid, by = "Species", suffix = c("", ".cen")) #A
dd centroid coordinates to ind_coords dataframe
ind_coords <- ind_coords %>% mutate( Axis1.cen = if_else(Species == "spec", Axis1, Axis
1.cen), Axis2.cen = if_else(Species == "spec", Axis2, Axis2.cen), Axis3.cen = if_else(Sp
ecies == "spec", Axis3, Axis3.cen) ) #Change axis
```

PCA plot specifications

```
n_pop <- nPop(data.genind)
if (n_pop <= 9) {
cols <- brewer.pal(max(n_pop, 3), "Set1")[1:n_pop]
} else {
# fallback for many groups
cols <- colorRampPalette(brewer.pal(9, "Set1"))(n_pop)
}
#Custom x and y labels
xlab <- paste("Axis 1 (", format(round(percent[1], 1), nsmall = 1), " %)", sep = "")
ylab <- paste("Axis 2 (", format(round(percent[2], 1), nsmall = 1), " %)", sep = "")

#Custom theme for ggplot2
ggtheme = theme(axis.text.y = element_text(colour="black", size=12),axis.text.x = elemen
t_text(colour="black", size=12), axis.title = element_text(colour="black", size=12), pan
el.border = element_rect(colour="black", fill=NA, linewidth=1), panel.background = eleme
nt_blank(), plot.title = element_text(hjust=0.5, size=15))

#Scatter plot axis 1 vs. 2
pca_plot <- ggplot(data = ind_coords, aes(x = Axis1, y = Axis2)) + geom_hline(yintercept
= 0) + geom_vline(xintercept = 0) + geom_segment(aes(xend = Axis1.cen, yend = Axis2.cen,
colour = Species), show.legend = FALSE) + geom_point(aes(fill = Species), shape = 21, si
ze = 3, show.legend = TRUE) + scale_fill_manual(values = cols) + scale_colour_manual(val
ues = cols) + labs(x = xlab, y = ylab) + ggtitle("Usnea PCA") + ggtheme

#Add ellipses, excluding 'spec'
ind_coords_for_ellipses <- ind_coords %>% filter(Species != "spec")

pca_plot_with_ellipses <- pca_plot + stat_ellipse( data = ind_coords_for_ellipses, aes(g
roup = Species, color = Species))
```

Plot PCA

```
print(pca_plot_with_ellipses)
```



Same result as PCA using PLINK – overlap between 2 species.